

Problem 26.2

Find an expression for the present value at time 0 of payments of \$55 at time 1 year, \$60 at time 2 years, \$65 at time 3 years and so on, up to the last payment at time 20 years.

Solution

The payments are

$$55, 60, 65, \dots$$

through time 20. This can be written as a level annuity of 50 plus an increasing annuity with payments

$$5, 10, 15, \dots, 100.$$

Thus, the present value is

$$50a_{\overline{20}|} + 5(Ia)_{\overline{20}|}.$$

For an increasing annuity-immediate,

$$(Ia)_{\overline{n}|} = \frac{a_{\overline{n}|} - nv^n}{i}.$$

Therefore,

$$(Ia)_{\overline{20}|} = \frac{a_{\overline{20}|} - 20v^{20}}{i}.$$

So the present value can be expressed as

$$PV = 50a_{\overline{20}|} + 5 \left(\frac{a_{\overline{20}|} - 20v^{20}}{i} \right).$$

Therefore,

$$\boxed{50a_{\overline{20}|} + 5(Ia)_{\overline{20}|}}$$

or equivalently,

$$\boxed{50a_{\overline{20}|} + 5 \left(\frac{a_{\overline{20}|} - 20v^{20}}{i} \right)}.$$